# Girls Build Bots — Brownie Coding Basics

## Badge-aligned workshop guide for robotics-team-led outreach

**Level:** Brownies (Grades 2–3) · **Badge:** Coding for Good — Coding Basics **Series:** Coding for Good (Coding Basics · Digital Game Design · App Development) **Format:** One 75–90 minute session · **Ratio:** 1 mentor per 3–4 Scouts **Who runs it:** A FIRST robotics team (FLL / FTC / FRC) leading the session. A physical robot is a nice bonus but **not required** — every activity runs on free browser tools and craft supplies the team or troop can provide.

> **Troop Leaders:** This curriculum is designed to *align with* the Brownie Coding for Good badge series. Official badge requirements are in your Girl Scout Volunteer Toolkit (VTK). Verify completed activities against the official steps before awarding badges. Badges are purchased through your council shop.

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## The through-line: coding *for good*

Every badge in this series ties coding to helping people. Brownies should leave believing that code isn't just for games — it's a tool they can use to make something better for someone else. Mentors: name this out loud. "We're not just coding — we're solving a problem for a real person."

**Vocabulary:** code, program, algorithm, bug, debug, sprite.

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## Coding Basics (75–90 minutes)

**Badge theme:** Discovering that code is a set of step-by-step instructions, and that algorithms shape the digital world around us.

### Materials

- Laptops or tablets with **ScratchJr** (tablet app) or **MakeCode Arcade** (arcade.makecode.com, any browser)
- Printable "algorithm cards" (simple picture steps: pour, stir, pour, etc.)
- A snack you can "program" the mentors to make (e.g., a cup of cereal + milk, or trail-mix scooping)

### Activity 1 — The Algorithm Snack (15 min)

An **algorithm** is just a list of steps in the right order. Scouts direct a mentor-robot to make a simple snack using only clear commands. Vague command ("make it good") → the robot freezes. Out-of-order command (milk before the cup) → a happy mess. **Debrief:** computers follow our steps *exactly*, so the order matters.

### Activity 2 — Unplugged Algorithm Cards (15 min)

In pairs, Scouts arrange picture cards into the correct sequence for an everyday task (brushing teeth, planting a seed). They "run" their partner's algorithm by acting it out and hunt for **bugs** — steps missing or out of order. First taste of debugging, no screen needed.

### Activity 3 — First Program On-Screen (35 min)

Mentor-led, one block at a time on a projector; Scouts mirror on their own device: - Make a character (a "sprite") - Make it move across the screen - Make it say hello or play a sound - Fast finishers: change the character and add a second action

Mentors circulate hands-behind-back, guiding with questions ("what do you think this block does?").

### Activity 4 — Where's the Code? (10 min) + Wrap (10 min)

Quick group talk: where is code hiding in their day? (traffic lights, video games, the microwave, a talking toy.) The point lands: algorithms are everywhere. Closing circle — each Scout shows one thing her program does.

> **Robot moment (optional):** Share for 60 seconds how the team's robot runs on code too — the same kind of step-by-step instructions, just controlling motors instead of a sprite.

### Completion Checklist — Coding Basics

By the end of the session, to earn the badge, each Scout can: - [ ] Explain that an algorithm is a list of steps in order - [ ] Put steps in the right sequence and spot a "bug" when they're wrong - [ ] Create and run a simple program with a sprite that moves or reacts - [ ] Name at least one place code shows up in everyday life

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## Mentor Guide (for the robotics team)

- **You don't need a robot for this.** It runs entirely on free browser tools and paper. If you brought a robot, the "robot moment" box is a 60-second add-on — not the main event.
- **Never touch the Scout's project.** Hands behind your back; guide with questions.
- **Celebrate bugs out loud** — "you found a bug, that's what real programmers do all day" — every single time.
- **Keep it small.** Brownies succeed when the goal is one sprite, one action. Rein in over-ambition gently.
- **Watch the clock.** Brownies fade after ~25 minutes on one thing; move to the next segment on schedule even if unfinished.
- **Name the "for good."** Say what problem the coding is solving and for whom. That's what separates this badge series from generic coding.

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## 📣 Share Your Impact — required for teams and troops using this curriculum

If you use Girls Build Bots materials to run a workshop, please complete all three:

1. **Credit us.** Tell your Scouts and audience that the curriculum comes from Girls Build Bots, and credit "Girls Build Bots" anywhere you share about the event.
2. **Tag us.** Post about your workshop and tag **@girlsbuildbots** on Instagram.
3. **Report your results.** Email **girlsbuildbots@gmail.com** with: the number of girls who participated, the number of badges earned, and a sentence or two of feedback on how it went.

This is how the program grows and how we measure our impact — your report becomes part of the Girls Build Bots story and helps the next troop and team.

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*Created for the "Girls Build Bots" Girl Scout Gold Award project. Original curriculum aligned to the Brownie Coding for Good badge series; official requirements © Girl Scouts of the USA, available to troop leaders in the Volunteer Toolkit.*